

Shahbaaz Mohammed

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Data Engineer

PROFESSIONAL SUMMARY:

- Experienced Data Engineer with over 5+ years of expertise in designing, developing, and optimizing scalable data pipelines, ETL workflows, and big data solutions across cloud and on-premise environments.
- Cloud-native expertise in AWS, GCP, and Azure, leveraging AWS S3, AWS Glue, AWS IAM, AWS Redshift, Snowflake, Google Cloud SQL, and PostgreSQL to architect and manage robust data infrastructures.
- ETL and data pipeline automation using Apache Spark, PySpark, Apache Airflow, Kafka, and AutoSys, ensuring efficient, real-time, and batch processing of terabytes of structured and unstructured data.
- Data warehouse modernization & migration, leading large-scale transitions from on-premises databases to Snowflake, HDFS, and AWS Redshift, optimizing performance and reducing operational costs.
- Optimized Spark and SQL workloads, leveraging partitioning, bucketing, caching, and indexing to enhance query performance and reduce processing time.
- Proficient in SQL, PostgreSQL, Python, Java, and Shell Scripting, developing high-performance queries, data transformations, and automation frameworks to streamline data processing and reporting.
- Experience in real-time data streaming and event-driven architectures using Apache Kafka, Spark Streaming, and AWS Kinesis, improving data availability and analytics capabilities.
- Built serverless ETL solutions using AWS Lambda and AWS Glue, reducing infrastructure maintenance overhead and improving data processing efficiency.
- Big Data ecosystem expertise, including Hadoop, HDFS, Hive, MapReduce, and Avro/Parquet file formats, optimizing storage and retrieval for high-volume data processing.
- Orchestrated machine learning data pipelines, integrating Apache Airflow, Databricks, and feature stores to support AI/ML model training and inference.
- CI/CD and DevOps practices, integrating Bitbucket, Git, Jenkins, Docker, and Kubernetes to ensure seamless deployment, version control, and infrastructure scalability.
- Agile & Scrum practitioner, actively contributing to sprint planning, daily stand-ups, retrospectives, and Jira tracking to align with business goals and ensure timely project delivery.
- Strong experience in modifying legacy Java applications, refactoring database connections, and rewriting SQL queries into HiveQL or Snowflake SQL, ensuring compatibility with modern data architectures.
- Data security & compliance expertise, implementing IAM policies, encryption strategies, and access controls in cloud environments to safeguard sensitive enterprise data.
- Collaborative team player, working closely with cross-functional teams, business stakeholders, and data analysts to drive data-driven decision-making and insights generation.
- Passionate about mentoring junior engineers, conducting knowledge-sharing sessions, and staying updated on emerging cloud, AI/ML, and data engineering trends.

TECHNICAL SKILLS:

- **Programming** : Python, Java, R, SQL, Shell Scripting
- **Databases & Data Storage** : MySQL, PostgreSQL, Oracle, Teradata, MS SQL, AWS S3, Google Cloud SQL, HDFS
- **Data Processing** : Databricks, Apache Spark, PySpark, AWS Glue, Apache NiFi
- **ETL & Data Integration** : Informatica, AutoSys, Talend, Apache Kafka, Airflow
- **Data Visualization** : Tableau, AWS QuickSight
- **Cloud Platforms** : AWS, GCP, Azure
- **Containerization** : Docker, Kubernetes
- **Security** : AWS IAM, Google Cloud IAM
- **CI/CD & Workflow Automation** : Jenkins, Git, Bitbucket
- **Agile Methodologies** : Agile, Scrum, Jira

PROFESSIONAL EXPERIENCE:**Client:** Charter Communications, St. Louis*Oct 2023 - Present***Role:** Data Engineer**Roles & Responsibilities:**

- Designed and maintained Python/PySpark-based ETL workflows to efficiently generate CRM summary data on a weekly basis.
- Led the migration of the legacy on-premises data warehouse to Snowflake, optimizing performance and enhancing cost efficiency.
- Designed and implemented scalable data pipelines using Apache Kafka, Spark and Python, processing millions of customer records daily, improving data flow efficiency.
- Managed AWS S3 for secure and scalable data storage, ensuring data availability and disaster recovery capabilities.
- Implemented AWS Glue to automate ETL processes, reducing manual efforts and improving data reliability.
- Implemented advanced SQL scripting for complex data manipulations and queries, while processing and analyzing millions of records in PostgreSQL
- Maintained AWS IAM configurations to manage secure access to AWS services, ensuring compliance with security policies.
- Created interactive dashboards and reports using AWS QuickSight, providing stakeholders with real-time insights and analytics.
- Automated data integration processes using Apache Airflow, optimizing workflow efficiency and reducing cycle times.
- Designed and executed data migration strategies to AWS Redshift, facilitating enhanced data analysis and storage.
- Contributed to Big data requirement analysis, and designed solutions for ETL and Business Intelligence platforms.
- Conducted performance tuning of Spark Applications, optimizing factors like Batch Interval time, Parallelism, and memory utilization.
- Develop and maintain Python-based applications to support enterprise-level Customer Experience solutions.
- Engineered real-time stream processing applications using Spark, Kafka, and Hive for streaming ETL and machine learning solutions.
- Implement and optimize batch processing workflows using Python, ensuring the accurate scheduling and execution of tasks in a Linux/UNIX environment.
- Applied Agile project management methodologies to ensure timely delivery of data projects, adapting to changing business needs.
- Conducted thorough unit and performance testing to validate system functionality and optimize scalability.
- Designed and implemented automated data quality checks using tools like Python and Apache Airflow, reducing datasets and streamlining the validation process.
- Collaborated with cross-functional teams to maintain consistency and accuracy in data processing and analysis across multiple platforms.

Environment: Python, Snowflake, HDFS, SQL, PostgreSQL Hadoop, Hive, Apache Airflow, Apache Kafka, AWS IAM, AWS S3, AWS Glue, AWS Redshift, PySpark, MapReduce, Oracle, HBase, Linux/Unix, AWS Quicksight, Tableau, Maven, Git, Agile, Jira.

Client: Tata Consultancy Services (TCS), India*June 2021 - July 2022***Role:** Application Developer**Roles & Responsibilities:**

- Led a team of four in the Omni Transformation project, focusing on migrating data from Teradata to HDFS.

- Analyzed the AutoSys Jobs, workflows and scripts of the legacy system to identify the jobs which need to be targeted for Migration.
- Proposed and implemented a new application Architecture, converting legacy systems from Teradata to HDFS, ensuring seamless integration and deployment.
- Utilized Sqoop to extract data from Teradata and transfer it to HDFS, optimizing the data transfer process and increasing throughput
- Redesigned the ETL processes using Informatica for data transformation, mapping, conversion, and loading, enabling smooth integration into the HDFS data lake.
- Performed Extract, Transform and load flat-file data into Target destination, ensuring smooth data insertion and updates with 99% accuracy and consistency across the new flow.
- Employed Avro, RC, and ORC file formats containing the Data from Teradata for data serialization, converting intricate objects into sequence bits.
- Developed DDL scripts to ensure accurate and efficient migration of tables to HDFS, preserving data integrity and optimizing query performance in the new environment.
- Designed and implemented PySpark scripts to automate data transfers from Teradata to HDFS, creating a seamless and efficient data pipeline that reduced manual intervention.
- Deployed the PySpark and Shell scripts in the Unix environments which are responsible for data transfer.
- Automated the execution of PySpark and shell scripts via AutoSys Jobs, ensuring scheduled data transfers to HDFS while maintaining data type compatibility and integrity.
- Modifying existing Java-based applications to replace JDBC connections from Teradata to Hive, enhancing application performance and system stability.
- Converted Teradata SQL queries with optimized HQL queries to replicate the exact table structure and maintain the same functionality within the new data flow, ensuring minimal disruption and system efficiency.
- Conducted extensive application testing to verify connectivity with HDFS, ensuring smooth data flow and reliable functionality post-migration.
- Established end to end connectivity of application and creating new AutoSys jobs to automate the application processes, maintaining system efficiency post-migration.
- Documented all development processes and code changes, adhering to business requirements and maintaining version control through Bitbucket for streamlined collaboration.
- Managed project tasks and sprints in Jira, ensuring timely delivery and seamless collaboration during the Teradata to HDFS migration.
- Contributed to the Agile process by actively participating in scrum meetings and sprint planning sessions, ensuring on-time project delivery and alignment with team goals.
- Worked closely with cross-functional teams, including QA Team, business analysts and other stakeholders to ensure successful execution of data migration.

Environment: Apache Hive, Oracle, MySQL, Teradata, Sqoop, Apache Hadoop, HDFS, MapReduce, Unix, Linux, Shell scripting, AutoSys, Informatica, Python, Java, Bit Bucket, Spark, Avro, Maven, Agile, Jira.

Client: Tata Consultancy Services (TCS), India

July 2019 - June 2021

Role: System Engineer

Roles& Responsibilities:

- Designed and implemented test strategies and plans for validating and testing data warehouse table structures and content, ensuring compliance with business and functional requirements across Hive, HDFS, and PostgreSQL.
- Developed and executed DDL scripts in HiveQL and PostgreSQL to create and manage tables in lower environments, ensuring correct schema design for testing and migration purposes.

- Managed lower environments (SIT, UAT) for effective testing, ensuring proper configuration and data flow using tools like AutoSys, Hadoop, and Hive.
- Optimized ETL pipelines to extract, transform, and load historical data from source systems (Teradata, PostgreSQL) to target data warehouse tables in HDFS and Hive using Spark and SQL.
- Developed automated ETL processes with Spark, Shell scripting, and Python, ensuring seamless data loading into target systems like HDFS, Hive, and PostgreSQL upon arrival of new data.
- Ensured high-quality data through rigorous validation and consistency checks, leveraging Hive and Spark to ensure data accuracy, completeness, and transformation integrity.
- Conducted in-depth data validation using complex SQL queries in PostgreSQL, Hive, and Spark to validate the source-to-target mappings and structural integrity of migrated data.
- Implemented automated testing solutions using Hadoop, Spark, and Unix shell scripts to enhance testing efficiency, minimize manual intervention, and ensure smooth data migration and processing.
- Documented defects, test results, and ETL processes, preparing detailed reports on data quality and project milestones using Bitbucket and Jira for version control and issue tracking.
- Actively participated in Agile methodologies, including sprint planning, daily stand-ups, and retrospectives, tracking project progress and tasks in Jira, ensuring continuous alignment with project timelines and business goals.
- Collaborated with cross-functional teams to ensure seamless data flow and testing across HDFS, Hive, and PostgreSQL.
- Provide mentorship and training in the use of relevant tools, software, and data warehousing processes to new team members.
- Contributed to knowledge-sharing sessions, leveraging tools like Bitbucket for code versioning and Jenkins for continuous integration, enhancing team capabilities and project delivery.

Environment: Informatica, PostgreSQL, Oracle, SQL, ETL, HDFS, Apache Hive, Spark, Sqoop, Hadoop, MapReduce, AutoSys, Bitbucket, Jenkins, Unix, Linux, Shell scripting, Python, Agile, Jira, Teradata.

Client: Tech Encryptors, India

Jan 2018 - June 2019

Role: Big Data/Hadoop Developer

Roles & Responsibilities:

- Designed and implemented test strategies and plans for validating and testing data warehouse table structures and content, ensuring compliance with business and functional requirements across Hive, HDFS, and PostgreSQL.
- Designing and implementing Big Data solutions that are scalable, high-performance, using Hadoop ecosystem technologies.
- Develop and optimize complex ETL processes that are used to ingest, transform, and analyze large volumes of data in the Data Warehouse.
- Design, maintain, and optimize Hadoop clusters using YARN ensuring optimal performance, reliability, and resource allocation.
- Employed efficient MapReduce, Hive, and Pig scripts and developed custom Hadoop components and plug-ins as needed.
- Implemented data security measures, ensuring compliance with privacy regulations, and establishing policies related to data governance.
- Optimize query performance and data storage strategies in HBase and other NoSQL databases like MongoDB.
- Developed and maintained data pipelines using Apache Kafka, also implementing real-time data streaming solutions with Apache Spark.
- Create comprehensive documentation for big data processes and architectures.

- Troubleshoot and resolving issues related to data processing, cluster performance, and job failures.
- Implementing data quality checks, processes for data cleansing, and metadata management strategies for ensuring data integrity.
- Develop and maintain data visualization dashboards using tools like Tableau.
- Develop automated testing frameworks for big data applications and ETL processes using Apache NiFi.
- Participating in daily stand-up meetings and agile development processes, contributing to team discussions on progress, challenges, and project planning.
- Stay updated about recent trends in the big data ecosystem and suggest enhancements to the existing systems.
- Contributed to knowledge-sharing sessions, leveraging tools like Bitbucket for code versioning and Jenkins for continuous integration, enhancing team capabilities and project delivery.

Environment: Java, Python, Apache Hadoop, HDFS, YARN, MapReduce, Hive, Pig, HBase, Apache Kafka, Spark, Apache NiFi, MySQL, MongoDB, DBMS, ETL, Tableau, Agile, Git, Jira.